

A SECURITY MODEL FOR CONTROLLER-PILOT DATA COMMUNICATION LINK

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Abstract

Communication systems in aviation tend to focus on safety rather than security. Protocols such as ADS-B are known to use plain-text, unauthenticated messages and thus are open to various attacks. Controller-Pilot Data Communication Link is no exception and was shown vulnerable also in practice. In this paper, we propose a cryptographic mechanism to provide secure mobility for CPDLC that can enable data encryption and authentication. The protocol is formally verified with the Proverif tool. We also estimate the byte overhead in CPDLC use.

I. Introduction

Air transport is an important enabler to achieve economic growth and development. A market report forecasts that air transportation is expected to at least double by 2050. Air Traffic Control (ATC) is the backbone of the world's air transport and as traffic load continues to grow dramatically [1], ATC has to handle more and more aircrafts.

In air traffic communication (ATC), typically, there are two types for communications mechanisms are being used between the pilot and controller. These mechanisms are (i) through voice communication (VHF) and (ii) through a datalink system. This last one, also called Controller pilot data link communications (CPDLC), has been added to avoid the overload of the voice communication system due to an increased amount of aircrafts, all using the same frequency bands. Both voice and data link will co-exist to organize the ATS (Air Traffic Service) communication, where CPDLC is mainly used in the case of non-time-critical communications (e.g., ATC traffic situation). Messages to be exchanged with CPDLC are, for instance, level assignments, crossing constraints, lateral deviations, route changes and clearances, speed assignments, radio frequency assignments, and various other requests. The message will show on the flight deck's visual monitor.

The controllers are authorized entities to issue ATC clearances (lateral deviations, level assignments, speed assignments, vectoring, etc.), radio frequency assignments, and numerous information demands. The pilots are also granted the freedom to respond to messages, inquire about receiving clearances and documents and record information. CPDLC is being implemented globally and it is currently in various stages of implementation. CPDLC is better compared to VHF voice due to the following mentioned reasons.

- 1) Unload ATC VHF-frequency communication.
- 2) Increased efficiency within the sector.
- 3) Pilot's requests can be handled simultaneously.
- 4) Reduced risk of miscommunication (e.g., confusion about required call sign).
- 5) Safer changes in frequency, thus less loss of communications events.

As the CPDLC enables communications between the ground and the air, it opens an immense attack vector between the controller and pilot. Interestingly, in [2] and [3], the authors shown that ADS-B and CPDLC are vulnerable to attacks. Such attacks may lead to life-threatening risks, and are safety alarming to the aviation industries. Therefore, the aeronautical telecommunications network also required a high level of security to protect their air traffic communication between pilots and controllers.

In this paper, we propose a lightweight cryptographic protocol to secure the CPDLC link. Compared to the prior work, it supports the handovers between different control zones. The proposed scheme exploits cryptographic mechanisms to provide secure key transfer between the ground and the air. We also perform formal verification of the protocol as well as estimate its computational overhead.

The rest of the paper is organized as follows. Section II gives the necessary background, Section III

explains related work, Section IV represents the network model, sections V and VI outline the cryptographic primitives and proposed solution, respectively. Security and performance analysis is done in Section VII. Finally, Section VIII deals with the future work and Section IX presents the conclusion.

II. Background

Air Traffic Management (ATM) is actively and continuously evolving since it is essential to provide direct contact channels between aircraft and controllers. This includes two distinct forms: voice-based and data link-based communication. Voice communication is commonly used in aviation, especially for ATC communications and is part of the aircraft's minimal equipment [4]. Most of this communication is carried out via VHF channels within the range 118.00 – 136.975 MHz, although some providers may use HF or SATCOM voice in remote locations.

With increasing air traffic, particularly around the major hub airports such as London Heathrow or Amsterdam Schiphol, the voice channels of VHF are getting extremely congested [5]. This has led to VHF channel spacing in Europe needing to be reduced from 25 kHz to 8.3 kHz. As a consequence, there is a push towards using data links for a variety of applications rather than voice.

In 1978, a specific purpose avionic data link was designed, named Aircraft Communications Addressing and Reporting Systems (ACARS), has been widely adopted since then and is now used for several purposes, with communications mainly free-text [4]. Over VDL2, SATCOM POA and HFDL links, these messages were transmitted.

Most of the network is operated in a similar manner by ACARS service providers to mobile cell networks for which subscribers pay a fee. With regards to ATC, the usage is identical to voice interaction but ATC over ACARS is not commonly used in practice. Although ACARS offers some ATC capabilities, there is no well-defined framework for managing the number of commands and exchanges required. Such a scheme is provided by the controller-pilot data link communications (CPDLC), allowing crew and controllers to share ATCs through data links. It is especially useful in congested areas and in Europe, where it has been deployed. From

2020, some areas of European airspace require the use of CPDLC [6].

The global aviation industry is progressing gradually towards CPDLC to improve voice communication between pilots and controllers. In the last 40 years, the business has depended on this. This does not mean that text messaging between controllers and pilots via Flight Management Systems (FMS) would substitute voice communications entirely, but there has to be a partial transition to handle the increase in air traffic. In the next two decades, this is projected to increase by 5% annually [7].

CPDLC security work has recently become more successful and correlates with its broader use. In addition, as this is carried on the normal, unprotected data links, there are possibilities for loss of integrity and issues of authenticity. The authors in [2] evaluated the threat to CPDLC from various types of attacks and discussed a number of possible cryptographic schemes to counter these threats.

III. Related Work

Safety has been a prevailing consideration in the aviation industry, resulting in it being built up around continuity and operational protection over the years. It is necessary to show the public that flying is safe by reducing accident hazards. Security is used to increase safety and is concentrated heavily on the prevention of physical attacks by implementing engaging systematic security checks, passenger screening and other stringent protective measures to enable connection to some areas and systems for air traffic control. The value of security policies intended to deter attacks will keep on rising where air traffic controllers are in-draft not to challenge the system's knowledge, as with the German ATC and the use of ADS-B [3].

Strohmeier [8] demonstrated that CPDLC is fundamentally insecure alongside other systems, as it imparts unauthenticated and unencrypted links of data easily eavesdropped and attackable. The authors [9] presented an interesting view of an experimental attack carried out in a controlled setting showing that the CPDLC system is susceptible to a man-in-the-middle attack using open source tools and applications. Though they allowed the intruder to enter the network physically, which the authors thought was unlikely to happen in real life, it indicates that the device is vulnerable on an application-level before it reaches the wireless format.

Eskilsson et al. [2] demonstrated the various attacks on ADS-B and CPDLC systems using software-defined radio (SDR). Varun et al. [10] pointed out the impact of faults in CPDLC and ADS-B systems. Gurtov et al. [11] discussed the CPDLC technical features and properties and created the corresponding model of threats. They also suggested some possible solutions to improve the security of data exchange used in air traffic management (ATM). Authors in [12] described symmetric and asymmetric cryptography schemes to secure the CPDLC. This paper also explained how cryptographic systems can be used to secure the exchange of all services for Air Traffic Services Communications (ATSC) and non-ATSC programs, including applications for Airline Operational Control (AOC).

International Civil Aviation Organization (ICAO) has specified in their draft [12] that guidance material on which Air Traffic Network (ATN) applications will promote peer and data traffic authentication as well as maintaining data confidentiality such that the information is not changed or duplicated. Authors [?] proposed a complete model for how to keep the ATN safe while reducing overhead protection. Likewise, in [13] the authors published a solution on how to secure CPDLC's authentication process using an elliptic curve-based protocol.

Griner [13] suggested an elliptic curve-based authentication protocol. The context management (CM) application is used during initial communication to handle mutual authentication and subsequent CPDLC application messages are authenticated using the coding scheme of ATN key message authentication. Their proposed protocol is based on

the elliptic curve digital signature algorithm (ECDSA) and it inherits the elliptical curve cryptosystems' security and implementation properties.

IV. Network Model and Attack Model

Network Model: Figure 1 shows a ground-to-sky communication model for the aviation networks. The aircraft communicates directly with the air traffic service units (ATSUs) receivers through the CPDLC (i.e., orange arrows), which uses a line-of-sight (LOS) basis.

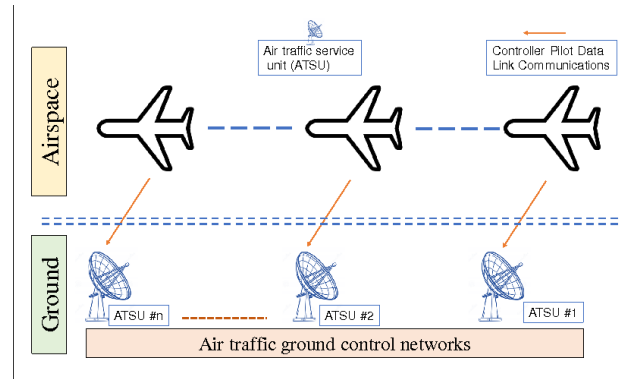


Figure 1. The Ground to Airspace Communication Network Model

Attack model: Consider a ground-to-sky communication model where the ATC is sending commands to the pilot up in the air. In such setting, assume that an attacker can take control of ground-to-air communication using some cheap devices. Table 2 describes a high level attack model on CPDLC-based ATC, as follows.

Table I. CPDLC Threat Model

Threat type	Affected Attributes	Actor type	Example Attack
Masquerading	Authentication, Non-Repudiation	Active	Ghost Aircraft/ Ground Station Identity
Injection	Availability, Confidentiality, Integrity, Non-Repudiation	Active	Aircraft Ghost Message/Ground Station
Eavesdropping	Confidentiality	Passive	Reading Messages
Alteration	Integrity	Active	Modification of Message Content

Eavesdropping is done when, without the consent of the communication parties, an unauthorized party listens to data traffic [14]. Since there is no current encryption in use for CPDLC, an attacker may perform an eavesdropping attack on CPDLC systems.

Injection is achieved by sending messages that are unauthorized. The data is said to be injected if it does not come from an authorized source. This attack can be quite serious and it is hard to detect if there is no necessary protection for the system to verify from which source the information is originally coming.

Alteration attack is specified when the attacker changes the legitimate data, being either modified, diverted, or delayed.

Masquerading means that the attacker impersonates and gains unauthorized privileges from an authorized user. The masquerading attacker has the means to engage in a complete conversation with its victim unless detected. An outsider may be able to issue instructions to the aircraft in the case of CPDLC.

V. Cryptographic Primitives

For the public key-based operations, we rely on the Elliptic Curve Cryptography (ECC) [15], offering more lightweight solutions than the classical RSA.

ECC is based on the algebraic structure of elliptic curves (ECs) over finite fields. We denote the curve in the finite field F_p with p a large prime by $E_{p(a,b)}$, defined by the equation $y^2 = x^3 + ax + b$ with a and b two constants in F_p and $\Delta = 4a^3 + 27b^2 \neq 0$. The base point generator of $E_{p(a,b)}$ of prime order q is denoted by P .

The EC multiplication $R = rP = (R_x, R_y)$ with $r \in F_q$ and $R_x, R_y \in F_p$ results in a point of the EC. The security of ECC is based on two computationally hard problems.

- The Elliptic Curve Discrete Logarithm Problem, this problem states that given two EC points R and Q of $E_{p(a,b)}$, it is computationally hard for any polynomial-time bounded algorithm to determine a parameter $x \in F_q^*$, such that $Q = xR$.
- The Elliptic Curve Diffie-Hellman Problem, given two EC points $R =$

$xP, Q = yP$ with two unknown parameters $x, y \in F_q^*$, it is computationally hard for any polynomial-time bounded algorithm to determine the EC point xyP .

The Schnorr signature scheme for message M is defined as follows. Assume that the sender S possesses the secret key pair (d_S, Q_S) with $Q_S = d_S P$ as the relation between private and public key. The public key Q_S is known by the receiver and verifier of the message. The sender first chooses a random value $r \in F_q^*$ and computes $R = rP$. Next, it computes $h = H(M \parallel R)$ and the actual signature $s = r - hd_S$. There are now three possibilities to be published:

- $\{M_S\}_{d_1} = \{M, R, s\}$
- $\overline{M}_{S_{d_1}} = \{R, s\}$
- $\tilde{M}_{S_{d_1}} = \{h, s\}$

Note that in the last two options, the message M is not included. This is because often a signature can be on parameters or information which is already known in advance and thus not need to be explicitly resend again. In case the variable R is sent (option 1 and 2), the receiver needs to compute $h = H(M \parallel R)$ and check if the equality $sP = R - hQ_S$ holds in order to verify the signature. In option 3, the receiver computes $R = sP + hQ_S$ and checks if $H(M \parallel R) = h$.

VI. Proposed Solution

We now describe how the aircraft can securely communicate with the ATSUs and securely establish a new symmetric secret session key (SK) during the handover as shown in Figure 2. This session key can then be used for the exchange of communication as long as the aircraft is in the operating zone of the ATSU. The usage of this key implicitly guarantees the mutual authentication of both entities. We distinguish three different phases: registration, initialization, and handover and describe the process between two ATSUs i.e, ATSU₁ and ATSU₂, and one aircraft. Note that ATSU₁ corresponds with the first ATSU to which the aircraft is communicating and takes over the role of ATSU₂ during the next handover process with another ATSU₃, who takes then the place of ATSU₂. The propose scheme is consists of registration phase, initialization phase and handover phase.

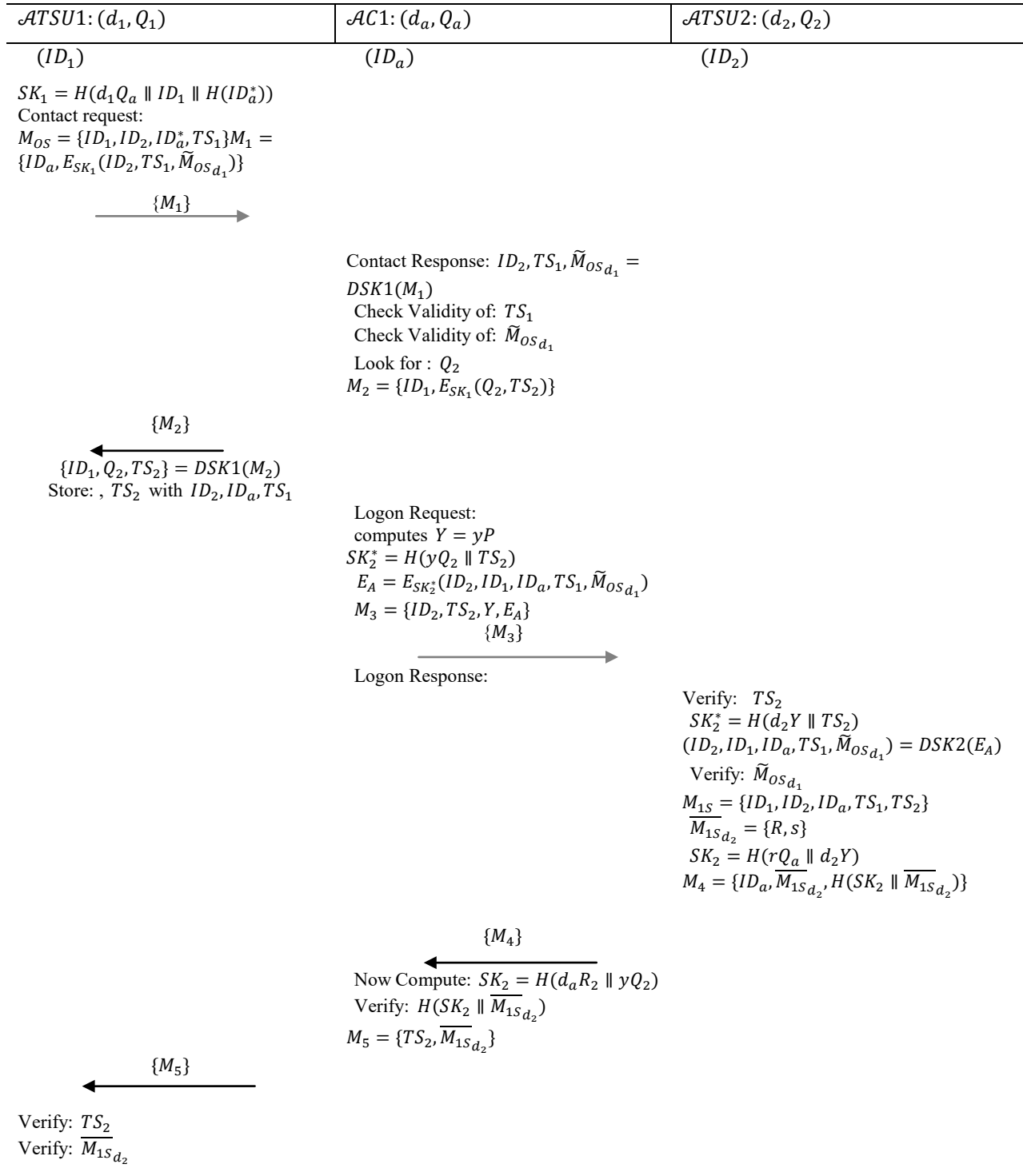


Figure 2. Proposed Authentication Scheme

1) *Registration phase*

a) *System set-up:*

Let us assume that each entity, being aircraft and ATSU, possesses a secret key pair, whose generation

is managed by a trusted third party (TTP). To be more specific, (d_1, Q_1) and (d_2, Q_2) for ATSU₁ and ATSU₂ are involved in the communication with the aircraft having the key pair (d_a, Q_a) . We denote the identities by ID_1, ID_2 for ATSU₁ and ATSU₂

respectively. Note that the identities of the ATSUs are represented by a four-character ICAO identifier, corresponding to 32 bits. Each ATSU also possesses the list of identities and public keys of their surrounding ATSUs.

b) Daily set-up:

Each day, all the ATSUs upload the list of identities, corresponding public keys and aircraft-related data like aircraft registration and/or aircraft address, departure and destination aerodromes of the aircraft active in its zone, during that day. We denote by ID_a the static identity of the aircraft and ID_a^* this additional information, which is unique every day. Similarly, the aircraft has installed the list of identities and corresponding public keys of the ATSUs on its route. In each list, corresponding certificates generated by a certificate authority (CA) are included in order to guarantee the connection between identity and public key [16].

2) Initialization phase

Before the aircraft leaves, it first establishes a session key with ATSU₁. This key is defined as $SK_1 = H(d_1 Q_a \parallel ID_1 \parallel H(ID_a^*))$.

3) Handover phase

We now explain the different steps to be performed by the different entities in the different steps of the handover phase.

- **Contact request:** Let TS_1 be the current timestamp. The ATSU constructs the message $M_{OS} = \{ID_1, ID_2, ID_a^*, TS_1\}$, generates a corresponding signature on it $\tilde{M}_{OS_{d_1}}$ and computes $M_1 = E_{SK_1}(ID_2, TS_1, \tilde{M}_{OS_{d_1}})$. The message $\{ID_a, M_1\}$ is sent to the AC. Note that SK_1 is the symmetric session key constructed in the previous key agreement, which is initially explained in the initialization phase and later defined in the logon phase for the next handover.
- **Contact response:** Upon arrival of M_1 , the aircraft being in the possession of SK_1 is able to decrypt the message and to retrieve $ID_2, TS_1, \tilde{M}_{OS_{d_1}}$. As a consequence, it can verify first the validity of the timestamp and then also the signature using Q_1 . If

positive, the aircraft looks up the corresponding public key Q_2 of ATSU₂. Next it constructs the message $M_2 = \{ID_1, E_{SK_1}(Q_2, TS_2)\}$, with TS_2 being the current timestamp, and sends it to ATSU₁, who can decrypt it and verify that the aircraft possesses the required info Q_2 for the handover. Moreover, it stores TS_2 together with TS_1, ID_2, ID_a until the next response of the aircraft is received.

- **Logon request:** To contact ATSU₂, the aircraft chooses a new random value y , computes $Y = yP$ and derives $SK_2^* = H(yQ_2 \parallel TS_2)$, $E_A = E_{SK_2^*}(ID_2, ID_1, ID_a, TS_1, \tilde{M}_{OS_{d_1}})$. The message $M_3 = \{ID_2, TS_2, Y, E_A\}$ is sent to ATSU₂.
- **Logon response:** ATSU₂ first checks the validity of the timestamp and then computes $SK_2^* = H(d_2 Y \parallel TS_2)$ in order to decrypt E_A and to be able to verify $\tilde{M}_{OS_{d_1}}$ after looking up the public key Q_1 belonging to the ATSU with identity ID_1 . If this signature is valid, it creates its own signature on the message $M_{1S} = \{ID_1, ID_2, ID_a, TS_1, TS_2\}$, resulting in $\overline{M}_{1S_{d_2}} = R, s$. Note that the random value r used in the signature generation to create $R = rP$ is also used to derive the new secret symmetric key with the aircraft, $SK_2 = H(rQ_a \parallel d_2 Y)$. Finally, the message $M_4 = \{ID_a, \overline{M}_{1S_{d_2}}, H(SK_2 \parallel \overline{M}_{1S_{d_2}})\}$ is sent to the aircraft.
- Upon arrival of M_4 , the aircraft can now compute $SK_2 = H(d_a R_2 \parallel yQ_2)$ and verify if $H(SK_2 \parallel \overline{M}_{1S_{d_2}})$ corresponds with the received value. If so, it further forwards $TS_2, \overline{M}_{1S_{d_2}}$ to ATSU₁, who looks up the info corresponding to the stored TS_2 and can thus verify the signature. If so, the connection with the aircraft stops and ATSU₂ takes over the communication.

VII Security and Performance Analysis

A. Formal Verification

The correctness of our scheme is also verified using the Proverif tool. This is one of the popular security verification tools and it uses the Dolev-Yao attack model. It can handle numerous cryptographic primitives, including shared and public-key cryptography (encryption and signatures), hash functions and key agreements with Diffie-Hellman. It can also verify the secrecy and authentication of the protocol.

We have two main channels for communication in our proposed security scheme, i.e., the private channel ChSec and the public channel ChPub. ChSec is a secure channel used for ATSU and aircraft

registration. Since our goal is to make a secure handover phase, we only focus on the handover phase security over the public channel in this research. ChPub is used for authentication of ID_1, ID_a and ID_2 . All participants compute and verify session keys. Hash, concatenation, encryption and decryption functions as constructors have already been declared. The ProVerif code demonstrates the performance of our proposed method in Figure 3. On the basis of the above-described results, it can be concluded that all three main processes of our protocol are successfully started and terminated. Results show that session keys are safe from any adversary attack. To summarize, using ProVerif, we are able to prove that our protocol for CPDLC handover preserves secrecy and attains secure authentication.

Completing equations... -- Process 1-- Query not attacker(SK1[]) in process 1 Translating the process into Horn clauses... Completing... Starting query not attacker(SK1[]) RESULT not attacker(SK1[]) is true. -- Query not attacker(xSK2[]) in process 1 Translating the process into Horn clauses... Completing... Starting query not attacker(xSK2[]) RESULT not attacker(xSK2[]) is true. -- Query not attacker(SK2[]) in process 1 Translating the process into Horn clauses... Completing... Starting query not attacker(SK2[]) RESULT not attacker(SK2[]) is true.	----- Verification summary: Query not attacker(SK1[]) is true. Query not attacker(xSK2[]) is true. Query not attacker(SK2[]) is true. -----
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Figure 3. Formal Verification Results Summary

B. Security Services Verification

1) Mutual Authentication

As mentioned in the registration phase, each entity, ATSU and airplane, is expected to know with which other entity there will be communication as both the identity and the corresponding public key are pre-stored before take-off.

First, the initial session key SK_1 can only be constructed by the aircraft and the ATSU of the airport due to the presence of a Diffie-Hellman key $d_1 Q_a$ in the hash construction of SK_1 .

Next, upon arrival of the contact request, the aircraft is able to verify the identity of the entity sending M_1 as it includes a signature of the ATSU. In addition, M_1 also includes information on the identity ID_2 of the next ATSU.

The authenticity of the contact response of the aircraft can then be verified by the ATSU as it is encrypted with the same initial session key SK_1 and includes data on the public key of ATSU₂, which can only be done in case the first contact request has been successfully decrypted.

Next, the aircraft starts the login request with the second ATSU, in which the initial signature of $ATSU_1$ on the message, including the identities of the involved entities, is included. Note that the message is encrypted by a temporary Diffie-Hellman key, only derivable by the intended ATSU.

Upon the arrival of the login request, the ATSU is the only one to be able to derive the actual message and thus to verify the initial message M_{OS} of $ATSU_1$. As a result, it also signs the message, including the involved entities and timestamps and hashes the result together with a Diffie-Hellman key constructed by means of the public key of the aircraft. As a consequence, the aircraft is the only one who can verify the authenticity and, after a positive verification, forwards the signature of $ATSU_2$ back to $ATSU_1$. Finally, $ATSU_1$ is able to verify the signature and thus be ensured of the authenticity of the involved entities during the process.

2) Message Integrity

Message integrity is achieved by using the Schnorr signature scheme. Assume that the sender S possesses the secret key pair (d_S, Q_S) with $Q_S = d_S P$ as the relation between private and public key. The public key Q_S is known by the receiver and verifier of the message. The sender first chooses a random value $r \in F_q^*$ and computes $R = rP$. Next, it computes $h = H(M \parallel R)$ and the actual signature $s = r - hd_S$.

$$\bullet \tilde{M}_{S_{d_1}} = \{h, s\}$$

The receiver computes $R = sP + hQ_S$ and checks if $H(M \parallel R) = h$.

3) Confidentiality

Confidentiality is one of the key requirements to avoid eavesdropping threats. The proposed system uses asymmetric cryptosystems to provide security, e.g., $M_1 = E_{SK_1}(ID_2, TS_1, \tilde{M}_{OS_{d_1}})$, $M_2 = \{ID_1, E_{SK_1}(Q_2, TS_2)\}$, $M_3 = \{ID_2, TS_2, Y, E_A\}$, $M_4 = \{ID_a, \overline{M_{1S_{d_2}}}, H(SK_2 \parallel \overline{M_{1S_{d_2}}})\}$ discussed in Section V. Every stakeholder has its own private key and identity. SK_1 is used to encrypt Q_2 , TS_2 and $\tilde{M}_{OS_{d_1}}$. Moreover, to decrypt $DSK1\{M_1\}$, the aircraft

must possess the corresponding key, otherwise, it cannot decrypt the garbled message.

C. Performance Evaluation

Our results show that the proposed security model is also well-suited for low-power devices (such as Raspberry PIs). The performance results are validated through the Raspberry PI-based testbed and other computer systems. The specs for our testbed are a 64-bit quad-core processor running at 1.4GHz processor and 1GB RAM. The specs for old and new PC are core-i5 and core-i7, having 3.3GHz and 2.2GHz processors with 4GB and 16GB RAM, respectively. For the evaluation, the Edwards-curve is used in the ECC algorithm. Furthermore, the concatenation is often used in auxiliary steps and denoted by a symbol .

Computational cost: Table II represents the average time for EC Addition, EC Multiplication and SHA 256 function for different hardware. In Table II the BouncyCastle library version 1.60 is used to implement cryptographic primitives for the Raspberry PI 3B+, Old and New PCs. To estimate the time needed by the above operations for the Raspberry PI 3B+, old and new PCs we used the Java Microbenchmark Harness (JMH) 1.21 toolkit.

Compared with Raspberry PI 3B+, PCs performed well in terms of EC Addition, EC Multiplication and digital signature. In a new PC, EC Addition took 4.86 microseconds approximately while in old PC and Raspberry PI 3B+ took approximately 18.90 and 156.80 microseconds, respectively which is slightly high compare to a new PC. Similarly, EC Multiplication is approximately done in 32800 microseconds in Raspberry PI 3B+, Old PC took around 4100 microseconds and a new PC took 1148.20 microseconds respectively. For the digital signature, a new PC took approximately 1.02 microsecond, while an old PC took approximately 1.69 microseconds. Similarly, Raspberry PI 3B+ took 13.53 microseconds approximately.

The computational complexity is depicted in Table III.

Table II. Execution Time

Device	Library	EC_Add (μs)	EC_Mul (μs)	SHA 256 (μs)	AES (μs)
Raspberry PI 3B+ Quad Core .4GHz, 1GB RAM	BouncyCastle	156.8 ± 0.7	32800 ± 20	13.52 ± 0.04	53.86 ± 0.17
Old PC Computer Intel Core i5-2500 CPU @ 3.3GHz x4, 4GB RAM	BouncyCastle	18.9 ± 0.5	4100 ± 100	1.693 ± 0.001	7.9 ± 0.2
New PC Computer Intel Core i7-8750H CPU @ 2.2GHz, 16GB RAM	BouncyCastle	4.866 ± 0.005	1148.2 ± 0.8	1.023 ± 0.002	2.90 ± 0.03

Table III. Computation Complexity

Operation	ATSU1	AC	ATSU2
Hash (H)	3H	4H	3H
Multiplication (M)	4M	6M	6M
Addition (A)	1A	1A	1A
Encryption (E)	1E	2E	0
Decryption (D)	1D	1D	1D
Total cost	3H+4M+1A+1E+1D	4H+6M+1A+2E+1D	3H+6M+1A+1D
Raspberry PI3B+	135.43 (ms)	203.08 (ms)	199.02 (ms)
Old PC	16.34 (ms)	24.64 (ms)	24.63 (ms)
New PC	4.6 (ms)	6.90 (ms)	6.90 (ms)

Communication cost: Table IV represents the size of identities, timestamp, hash functions and EC curve, respectively. In the proposed scheme, we have 5 messages and each message has a different length. For instance, the length of message M_1 is 608 bits. For message M_2 , the communication cost is 320 bits. Likewise, the communication cost for messages M_3 and M_4 are 960 and 800 bits, respectively. Finally, message M_5 has a communication cost of 544 bits.

Table IV. Memory Consumption

IDs (bits)	TSs (bits)	Hash Function (bits)	EC Curve (bits)
32	32	256	256

VIII. Future Work

In this paper, we proposed several cryptographic mechanisms to provide secure key transfer for CPDLC that can enable data encryption

and authentication. Our proposed solution for CPDLC solved the most important security issues present in CPDLC. In order to provide better availability protection, we will have to develop in addition a load balancer at the central communication node.

The security of another protocol used in aircraft communication, being Automatic dependent surveillance-broadcast (ADS-B), is also becoming a considerable concern due to the current way the flight system broadcasts essential and detailed information about their aircraft, their location, speed, altitude, and other data over links and networks that are not encrypted. These current practices allow hackers and intruders easy access to launch jamming, eavesdropping and message modification attacks on the aircraft in flight. In the future, we intend to also create a system where confidentiality, integrity and availability (CIA) will be obtainable with the broadcasting of message, availability of

authentication, message detection by IDS on the flight and the use of spider monkey approach to stop intruders.

IX. Conclusion

Controller Pilot Data Communication (CPDLC) is currently unprotected and vulnerable to attacks but mandatory to use for pilots. We proposed a lightweight cryptographic protocol to secure the link. Compared to the prior work, it supports the handovers between different control zones. We performed a formal and informal verification of the protocol as well as calculated its computational overhead.

References

- [1] E. Mazareanu, Transportation logistics, Available at <https://www.statista.com/statistics/193533/growth-of-global-air-traffic-passenger-demand/> (11.05.2020).
- [2] S. Eskilsson, H. Gustafsson, S. Khan, and A. Gurtov, "Demonstrating ADS-B and CPDLC Attacks With Software-Defined Radio," in 2020 Integrated Communications Navigation and Surveillance Conference (ICNS), IEEE, 2020, 1B2–1.
- [3] M. Strohmeier, M. Schäfer, R. Pinheiro, V. Lenders, and I. Martinovic, "On Perception and Reality in Wireless Air fieraCommunication Security," IEEE Transactions on Intelligent Transportation Systems, vol. 18, no. 6, pp. 1338–1357, 2016.
- [4] C. Spitzer, U. Ferrel, and T. Ferrell, Digital avionics handbook. CRC Press, 2015.
- [5] M. Olive, "Efficient Datalink Security in a Bandwidth-Limited Mobile Environment-an Overview of the Aeronautical Telecommunications Network (ATN) Security Concept," in 20th DASC. 20th Digital Avionics Systems Conference (Cat. No. 01CH37219), IEEE, vol. 2, 2001, 9E2–1.
- [6] Eurocontrol, Don't Text While Driving. Text While flying! Why Use Controller -Pilot Data Link Communications, Available at <https://www.eurocontrol.int> (17.12.2020).
- [7] B. Woodrow, Tcpldc: Transitioning to Controller Pilot Data Link Communications, Available at [https://www.aviationtoday.com/2014/12/01/tcpldc-transitioning-to-controller-pilot-data-link-](https://www.aviationtoday.com/2014/12/01/tcpldc-transitioning-to-controller-pilot-data-link-communications/) communications/ (11.05.2020).
- [8] M. Strohmeier, "Security in Next Generation Air Traffic Communication Networks," PhD thesis, University of Oxford, 2016.
- [9] D. Di Marco, A. Manzo, M. Ivaldi, and J. Hird, "Security Testing with Controller-Pilot Data Link Communications," in 2016 11th International Conference on Availability, Reliability and Security (ARES), IEEE, 2016, pp. 526–531.
- [10] V. S. Sudarsanan, M. A. Jacobs, A. Dervisevic, and D. DeLaurentis, "ADS-B and CPDLC Fault Modeling for Safety Assessment in a Distributed Environment," in 2018 IEEE Aerospace Conference, IEEE, 2018, pp. 1–14.
- [11] A. Gurtov, T. Polishchuk, and M. Wernberg, "Controller-Pilot Data Link Communication Security," Sensors, vol. 18, no. 5, p. 1636, 2018.
- [12] T. McParland, V. Patel, and W. Hughes, "Securing Air-Ground Communications," in 20th DASC. 20th Digital Avionics Systems Conference (Cat. No. 01CH37219), IEEE, vol. 2, 2001, 7A7–1.
- [13] D. Getachew and J. H. Griner, "An Elliptic Curve Based Authentication Protocol for Controller-Pilot Data Link Communications," in Integrated CNS Technology Conference Workshop, 2005.
- [14] H. Imai, Wireless Communications Security. Artech House, 2005.
- [15] N. Koblitz, "Elliptic Curve Cryptosystems," Mathematics of Computation, vol. 48, no. 177, pp. 203–209, 1987.
- [16] V. Patel and T. McParland, "Public Key Infrastructure for Air Traffic Management Systems," in [17] 20th DASC. 20th Digital Avionics Systems Conference (Cat. No. 01CH37219), IEEE, vol. 2, 2001, 7A5–1.

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